

Neurosymbolic Computing

Summer Semester 2026

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Data Science Group
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Outline

1 Organization

2 Presentation and Report

3 Timeline

4 Topics

Overview

- Advisors:
 - Prof. Dr. Michael Gertz (gertz@informatik.uni-heidelberg.de)
 - Jonas Gann (gann@informatik.uni-heidelberg.de)
 - Marina Walther (walther@informatik.uni-heidelberg.de)
- Time and place:
 - Thursdays, 2–4pm (c.t.)
 - INF 205 (Mathematikon), seminar room 1
- Office hours:
 - Prof. Dr. Michael Gertz: Thursdays 1–2pm (or by appointment)
 - Further information on our group's website at <https://ds.ifi.uni-heidelberg.de/>

Overview

- Audience:
 - Bachelor and master students in Computer Science, Scientific Computing and related programs of study
- Suggested knowledge:
 - Main principles and methods underlying transformers (e.g., ChatGPT or Claude), chatbots, prompting; basic knowledge of natural language processing (NLP) and machine learning will be helpful.
 - Basic knowledge of logics and logic programming (e.g., Prolog, Datalog, or Lean) is recommended.
 - See also “Required and suggested reading” on seminar web page in Moodle.

Requirements

How to get the ECTS:

- Presence at all presentations (no absence without valid excuse)
- Read **mandatory reading** papers (see Moodle) prior to the first presentation
- Read through the presentation “How to give a presentation” (see Moodle)
- Give your presentation based on a **collection** of papers
(30 minutes, plus 10 minutes discussion)
- Read papers related to one other presentation and prepare and ask questions
⇒ **we want to have lively discussions!**
- Hand in report
- Final grade: presentation (50%), questions (8%), and report (42%)
- Presentation and report each have to be at least satisfactory (4.0), otherwise the seminar requirements won't be satisfied (grade = 5.0, not passed)

Schedule

Preliminary schedule: (depends on the final number of students)

- today: preliminary discussion
- until Monday, April 20, 8pm: selection of topic(s) by students
- by Wednesday April 22: assignment of topics by advisors
- Starting May 21: presentations, with two presentations per session

Moodle

All **information and materials, incl. papers**, are available via:

<https://moodle.uni-heidelberg.de/course/view.php?id=31555>
(these slides are also available via Moodle!)

- Seminar *Neurosymbolic Computing*
(listed under “Fakultät für Mathematik und Informatik / Informatik / Seminare”)
- Registration via Moodle is required!
 - Enrollment key: NeuroSym26HD
- **Note:** You need to enroll via heiCo once you have been assigned and agreed to a topic!

If you cannot access articles from home:

- login from campus or use the university VPN
<https://www.urz.uni-heidelberg.de/de/vpn>

Presentation: Preparation

- **Mandatory preliminary discussions with advisors**
(as early as possible but no later than 2 weeks before your presentation)
 - Clarification of open questions
 - Layout of the plan for your presentation
- **1 week before your presentation**
 - Submit preliminary slides as PDF
 - Template for slides can be found on Moodle course page
- **No later than three days before presentation**
 - Submit final slides as PDF to your supervisor for upload in Moodle
- **Duration of presentation:** 30 minutes + 5-10 minutes questions and discussion;
30 minutes is a hard constraint!
Also: **No reading from cue cards, script, or laptop; talk to the audience!**

Not complying with the above will have a negative impact on your final grade!

Question Session

- Following each presentation, we want to have a lively discussion about the methods, techniques, applications, and assumptions that have been presented.
- For each presentation, one student will prepare at least two critical, insightful, meaningful questions related to the paper(s) content and contributions that have been presented.
- For this, the student asking questions has to read respective paper(s) as well.
- This does **not** mean that everyone else is not allowed or not expected to participate in the discussion!

Report: What is expected?

On Moodle class pages

- Template for report
- contains many points and hints, especially regarding standards of scientific writing
- read this report carefully, as many aspect apply to your thesis as well!

Submission of Report

- Submission on or before August 6, 2026, 8pm.
- Hints:
 - Keep your final exams in mind, start early with the report
 - Start writing (and complete) your report as soon as possible after your presentation
 - You are welcome to submit your report before the deadline
- No return of report for further corrections
 - ⇒ poor report, poor grade
- Submission of your report: send the signed PDF to your advisor

Selection of Topics

- Register asap for this seminar in Moodle
- Get an overview of the topics
- Look at / skim through the papers you are interested in the most
- Fill in the questionnaire “Selection of Topics” in Moodle
- Submit information until Monday, April 20, 8pm
- Assignment of papers/topics to students by Wednesday, April 22
- The questionnaire asks for the following information:

Selection of Topics /2

- 1 Your name
- 2 Your program of study, semester of study, matriculation number
- 3 What other courses offered by our group have you already taken?
- 4 Is this your first seminar?
- 5 Do you need this seminar? For example, is this the last seminar you need in order to complete degree requirements?
- 6 Ranked top-3 list of topics/papers you would like to work on
- 7 2-3 sentences stating why you are interested in the topics you selected (and how they are related to possible subsequent projects such as a bachelor's or master's thesis)

Note: we will also use your top-3 list to determine the topic/paper(s) you are supposed to prepare questions for.

Preliminary Schedule

Schedule for the seminar

April 16	preliminary discussion
April 20, 8pm	deadline for top-3 wish-list
April 22	Assignment of topics
May 21	first presentations (1+2)
June 28	presentations (3+4)
June 11	presentations (5+6)
June 18	presentations (7+8)
June 25	presentations (9+10)
July 2	presentations (11+12)
July 9	presentations (13+14)
August 6	deadline for submitting report

Topics – See extra slides

Thank you for your attention!

Questions?